

EASTERN INTERNATIONAL UNIVERSITY
SCHOOL OF COMPUTING
AND INFORMATION TECHNOLOGY



Practice Assignment – Quarter 1, 2025-2026
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Course Code: CSW 309
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LAB 02

Topic: Identify actors, define use cases, and draw the Use Case Diagram for the Equipment Borrowing Management System

1: Identify actors and the system boundary based on the Lab 01 scenario.

i. Actor:

Actor	Description	Main Goals / Interactions
Borrower (Teacher / Student)	A user who needs equipment for teaching or learning activities	Search for available equipment, place borrowing requests, view request progress, return equipment after use
Approver / IT Staff	Person responsible for managing equipment allocation	Review and approve requests, adjust schedules if there is a conflict, confirm returned items, update equipment status
Administrator	System manager controlling overall operations	Maintain user roles, manage system configurations, generate analytics reports
Notification Service	External module or system component	Automatically sends alerts such as request updates or due-date reminders
System Timer (Scheduler)	Background automation mechanism	Periodically checks overdue items and triggers reminder workflows

ii. System Boundary

The EBMS supports the entire lifecycle of borrowing and returning teaching equipment.

Functional Capabilities Included in System Scope

Online borrowing request submission
Equipment approval workflow with scheduling
Equipment availability lookup and real-time status tracking
Automatic notifications (request status, reminders, overdue warnings)
Reporting dashboard and usage analytics
Role-based authentication and authorization
Secure data storage and activity logging

2: List and group business use cases according to each actor's goals.

i. Actor: Borrower (Teacher / Student)

Business Goals: Borrow and manage equipment for teaching/learning activities

Goal Group	Use Case
Access & Authentication	UC01 – Login / Logout
Borrowing Operations	UC02 – Search equipment UC03 – Submit borrowing request
Monitoring Status	UC04 – Check request progress UC05 – View borrowing history
Adjust Borrowing Plan	UC06 – Cancel a pending request UC07 – Request borrowing time extension
Maintenance Reporting	UC08 – Report damaged/malfunctioning items
Return Process	UC09 – Return equipment

ii. Actor: IT Staff (Approver)

Business Goals: Ensure proper allocation & safe device return

Goal Group	Use Case
Request Management	UC10 – View all requests UC11 – Approve request UC12 – Reject / Reschedule request
Inventory & Status Maintenance	UC13 – Update equipment condition UC14 – Confirm returned equipment UC15 – Mark lost or heavily damaged items
Operational Tracking	UC16 – Monitor overall loan process & workload

iii. Actor: Administrator

Business Goals: Govern system operations and data integrity

Goal Group	Use Case
User & Role Management	UC17 – Manage user accounts & permissions
Equipment Database Management	UC18 – Manage equipment catalog
Policy & Configuration	UC19 – Define borrowing rules UC20 – Configure notification preferences

Monitoring & Reporting	UC21 – View action logs UC22 – Generate usage reports & analytics
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iv. Actor: Notification Service (External System)

Business Goals: Automated communication to stakeholders

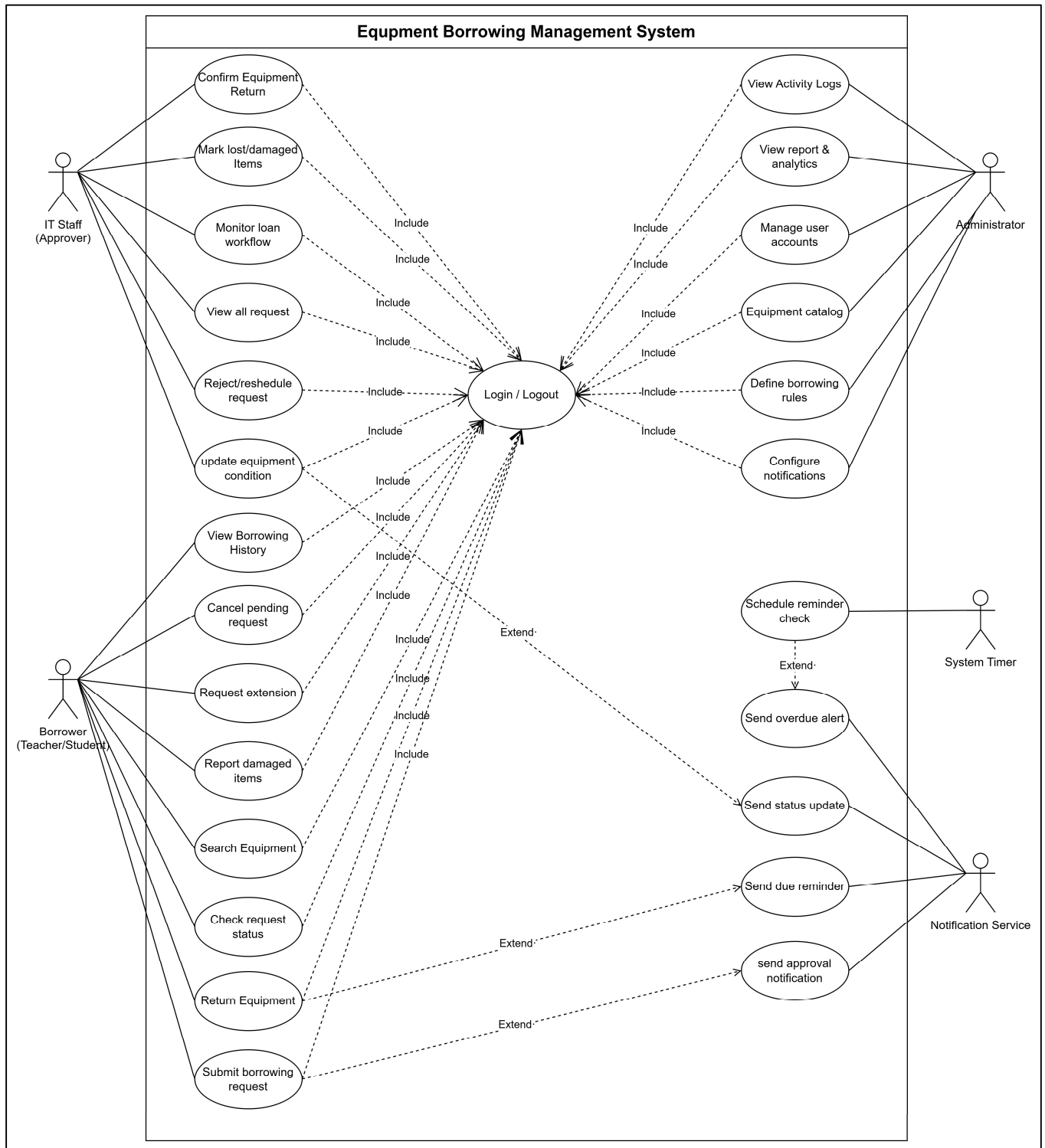
Use Case
UC23 – Send approval/rejection updates
UC24 – Send due-date reminders
UC25 – Send overdue notices
UC26 – Send equipment condition/status alerts

v. Actor: System Timer (Scheduler)

Business Goals: Handle background automated tasks

Use Case
UC27 – Execute scheduled reminder & overdue checks

3: Model a Use Case Diagram (including include/extend relationships when appropriate).



4: Write fully dressed use case specifications for 2–3 key use cases (Borrow Equipment, Approve Request, Return Equipment).

i. Borrow Equipment (Submit Borrowing Request)

Field	Description / Content
Use Case ID / Name	UC03 – Submit Borrowing Request

Scope / System	Equipment Borrowing Management System
Level	User Goal
Primary Actor	Borrower (Teacher / Student)
Supporting Actors	IT Staff, Notification Service
Stakeholders & Interests	Borrower – wants to reserve needed equipment quickly; IT Staff – wants accurate, complete requests; Admin – wants a smooth approval workflow.
Goal in Context	Allow users to request equipment online for specific time, purpose, and location.
Preconditions	Borrower is authenticated; equipment is available in inventory.
Trigger	Borrower clicks “Request Equipment” on the system.
Main Success Scenario (Basic Flow)	1. System displays the borrowing form. 2. Borrower fills in request details (equipment, date, purpose). 3. System validates information and checks availability. 4. Borrower submits the request. 5. System records it as Pending . 6. System notifies IT Staff and sends confirmation to Borrower.
Extensions (Alternate Flows / Exceptions)	3a. Equipment not available → System suggests next available date. 2a. Missing or invalid fields → System prompts correction. 5a. Network issue → Request temporarily stored in local cache until connection resumes.
Postconditions / Guarantees	Request saved as Pending with unique ID; Borrower and IT Staff are notified.
Special Requirements (NFR)	Form submission < 3 seconds; must validate every required field; accessible on desktop and mobile.
Technology / Data Variations	Borrowing form fields may vary between teacher and student.
Assumptions	Internet connection stable; user has valid system credentials.
Notes / Issues	Future upgrade: auto-check conflicts across departments.

ii. Approve Request

Field	Description / Content
Use Case ID / Name	UC11 – Approve Request
Scope / System	Equipment Borrowing Management System
Level	User Goal

Primary Actor	IT Staff
Supporting Actors	Notification Service
Stakeholders & Interests	IT Staff – needs accurate info to approve fairly; Borrower – expects timely response; Admin – monitors system consistency.
Goal in Context	Review and decide whether to approve, reject, or reschedule borrowing requests.
Preconditions	At least one request is pending approval.
Trigger	IT Staff selects a pending request from the list.
Main Success Scenario (Basic Flow)	1. System shows list of pending requests. 2. IT Staff reviews details (equipment, date, borrower info). 3. System checks schedule and availability. 4. IT Staff selects Approve, Reject, or Reschedule . 5. System updates request status. 6. Notification Service alerts the Borrower with the result.
Extensions (Alternate Flows / Exceptions)	3a. Equipment unavailable → System offers next available slot. 2a. Request missing details → IT Staff returns for borrower revision. 5a. Approval error (conflict detected) → System prevents saving until resolved.
Postconditions / Guarantees	Request status updated; borrower notified; system logs decision.
Special Requirements (NFR)	Changes reflected instantly; decision logged for audit.
Technology / Data Variations	Interface may display a calendar or equipment timeline view.
Assumptions	IT Staff has sufficient permissions.
Notes / Issues	Optional enhancement: automated approval for repeat borrowers.

iii. Return Equipment

Field	Description / Content
Use Case ID / Name	UC09 – Return Equipment
Scope / System	Equipment Borrowing Management System
Level	User Goal
Primary Actor	Borrower (Teacher / Student)

Supporting Actors	IT Staff, Notification Service, System Timer
Stakeholders & Interests	Borrower – wants quick confirmation of return; IT Staff – ensures condition check; Admin – tracks device lifecycle; Notification Service – sends reminders.
Goal in Context	Record the return of borrowed equipment and update system availability.
Preconditions	Equipment is currently borrowed and not yet marked as returned.
Trigger	Borrower selects “Return Equipment” or physically returns the item.
Main Success Scenario (Basic Flow)	1. Borrower initiates a return request. 2. IT Staff inspects equipment and confirms condition. 3. System updates status to Returned . 4. System Timer cancels overdue tracking. 5. Notification Service sends confirmation to Borrower.
Extensions (Alternate Flows / Exceptions)	2a. Item damaged → IT Staff logs issue and updates repair report. 3a. Item not found → System flags as “Missing” and alerts Admin. 1a. Borrower fails to return on time → System Timer triggers overdue alert.
Postconditions / Guarantees	Equipment status = “Available”; all return actions logged; borrower notified.
Special Requirements (NFR)	Return confirmation visible instantly; response time ≤ 2 seconds.
Technology / Data Variations	Return page may include checklist or photo upload for proof.
Assumptions	IT Staff available for inspection; stable connection.
Notes / Issues	Future upgrade: self-return kiosk integration with QR code scan.